

Tuesday -Week 3- Solution

Q. 1 — Explain Workflow orchestration

Question

Explain the concept in plain language. Include:

- its main purpose;
- where it fits in the lesson workflow;
- one mistake a new learner should avoid.

Ans - Orchestration decides what should run, when it runs.

A useful way to place it is to connect it to the input, processing step, and expected output described in the lesson. A common mistake is treating the concept as an isolated definition instead of checking its behaviour in the wider workflow.

Q. 2 — Connect Workflow orchestration and Apache Airflow

Question

- Explain the role of each concept.
- State how they are related.
- Identify one reason they should not be treated as interchangeable.

Workflow orchestration : it decides what should run, when it should run, and what is allowed to run next. Without orchestration, a team might start scripts manually or use separate timers.

Apache Airflow: Airflow is a platform for defining, scheduling, monitoring, and retrying workflows. A workflow is written as a DAG.

They are related because the first concept establishes part of the context or capability used by the second. They are not interchangeable: each has a different responsibility, input, or decision point.

MCQ : Q. 5 Purpose of Workflow orchestration
Which statement best matches the lesson?

Ans - A. In plain words: Orchestration decides what should run, when it should run, and what is allowed to run next. Without orchestration, a team might start scripts manually or use separate timers.

Q. 6 — MCQ: Applying Apache Airflow

What should the learner do before accepting a result involving **Apache Airflow**?

Ans - B. Verify the output against the requirement and retain evidence.

Q. 11 — MCQ: Purpose of Scheduler

Which statement best matches the lesson?

Ans - A. The scheduler decides when DAG runs should be created and which task instances are ready to run. If the webserver is running but the scheduler is not, the UI may open but scheduled.

Q. 12 — MCQ: Applying Webserver and Airflow UI

What should the learner do before accepting a result involving **Webserver and Airflow UI**?

Ans - B. Verify the output against the requirement and retain evidence.

Q. 17 — MCQ: Purpose of Executor

Which statement best matches the lesson?

Ans - A. The executor controls how tasks are launched. For beginners, remember the responsibility rather than memorising every executor type: The scheduler decides what is ready.

Q. 18 — MCQ: Applying DAG

What should the learner do before accepting a result involving **DAG**?

Ans - B. Verify the output against the requirement and retain evidence.

Q. 23 — MCQ: Purpose of Operator

Which statement best matches the lesson?

Ans - A. An operator is a template used to create a task. Core distinction: Operator = task template.

Q. 24 — MCQ: Applying DAG file structure

What should the learner do before accepting a result involving **DAG file structure**?

Ans - B. Verify the output against the requirement and retain evidence.